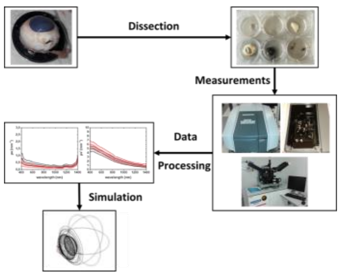
Determination of Optical Parameters of the Porcine Eye and Development of a Simulated Model

Simon REGAL,[1] Denise O'CONNOR,[2] Pauline BRIGE,[3,4] Roger DELATTRE,[1] Thierry DJENIZIAN,[1] Marc RAMUZ,[1,]*

[1] Department of Flexible Electronics, Ecole Nationale Supérieure des Mines, Centre Microélectronique de Provence CMP-EMSE, MOC, F-13541 Gardanne, France.
[2] Department of Bioelectronics, Ecole Nationale Supérieure des Mines, Centre Microélectronique de Provence CMP-EMSE, MOC, F-13541 Gardanne, France.
[3] LIIE, Aix Marseille Univ, Marseille, France
[4] CERIMED, Aix Marseille Univ, Marseille, France.

Key words: spectroscopy, porcine ocular tissues, optical properties, eye simulation model

The eye is a very sophisticated system of optical elements for the preeminent sense of vision. In recent years, the number of laser surgery to correct the optical aberration such as myopia or astigmatism has significantly increased. Consequently, improving the knowledge related to the interactions of light with the eye is very important in order to enhance the efficiency of the surgery. For this reason, a complete optical characterization of the porcine eye is presented in this study. Kubelka-Munk and Inverse Adding-Doubling methods were applied to spectroscopy measurement to determine the absorption and scattering coefficients. Furthermore, the refractive index has been measured by ellipsometry. The different parameters were obtained for the cornea, lens, vitreous humour, sclera, iris, choroids and eyelid in the visible and infrared region. Thereafter, the results are implemented in a COMSOL Multiphysics® software to create an eye model. This model gives a better understanding of the propagation of light in the eye by adding optical parts such as the iris, the sclera or the ciliary bodies. Two simulations show the propagation of light from the cornea to the retina but also from the sclera to the retina. This last possibility provides a better understanding of light propagation during eye laser surgery such as for e.g. transscleral cyclophotocoagulation.



Eye simulation models allow the development of new laser treatments in a simple and safe way for patients. To this purpose, the creation of an eye simulated model based on optical parameters obtained from experimental data is presented in this study. This model will facilitate the understanding of the light propagation inside the porcine eye.

This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process, which may lead to differences between this version and the Version of Record. Please cite this article as doi: 10.1002/jbio.201800398

Introduction

The total number of laser eye surgeries increased by 6.2% between 2017 and 2018, more than 843,000 were performed in 2018 in the United States.[1] They are performed to correct various ophthalmological problems such as myopia, presbyopia, hyperopia or astigmatism or to treat glaucoma. The eye is a complex organ presenting various optical properties because of its different layers of tissues. Some parts are transparent, including the cornea, the lens, the vitreous humour, characterized by a low absorption and scattering coefficients, whereas the choroids, the ciliary body, the sclera and the eyelid absorb and diffuse the light.[2] Because of these differences, light propagation inside the eye is difficult to precisely address. However, absorption and scattering coefficients are important to create a simulated model in order to increase precision of optical coherent tomography as well as laser surgery.

Light propagation in a turbid medium is characterized by refractive index (n), anisotropic factor (g), absorption (μa) and scattering (μs) coefficients.[3] Research groups have reported different results for these parameters. The refractive index of the cornea, lens, aqueous and vitreous humour are well known for the human eye.[4–6] Furthermore, the refractive index of the lens is characterized by a non-uniform gradient, greater in the centre than in the periphery. Gullstrand[4] and Liou and Brennan[6] have developed mathematical expressions of this gradient. Some studies also provided the value for porcine lens.[7–9] Concerning the aqueous and vitreous humour, results are available for porcine[10], bovine[11] and human.[12] In the same way, the value for the sclera is calculated[13] and measured for human.[14] No data were found concerning the choroids, the iris and the eyelid.

Research groups have also reported transmission (T) and reflection (R) spectrums for the whole human eye[15,16] but mainly for specific parts.[17–20] However, these results are not generalizable to create a model because the transmission and reflection spectrum depend on the thickness of media. The absorption and scattering coefficients are extracted from the transmission and reflection spectrum, ignoring the dependency on thickness. The absorption and scattering coefficients of the cornea[21,22], the lens[21,22], the vitreous humour[11], the sclera[13,14,23–25], the iris[26] and the choroids[23] are also discussed in the literature.

Overall, we found few consistent values concerning the refractive index, the absorption and scattering coefficients in the literature. Moreover, research groups have worked on specific parts of the eye for one or more defined wavelengths. Although, these data cover the optical properties of the eye, the researchers typically do not use the same equipment, methods or even animals to calculate coefficients. For example, some groups made their own integrated sphere[25] or only studied specific wavelengths[21,22,25]. Similarly, studies are often carried out on porcine, rabbit, bovine and human tissues. Thus, the poor consistency of results as well as the lack of data for some eye tissues has motivated this work.

The objective of our work is to systematically determine the absorption coefficient, the scattering coefficient and refractive index of porcine ocular tissues from the visible to the near-infrared region. The choice of the porcine eye is due to the functional and anatomical similarity with the human eye[27]; as well the difficulty of obtaining human eyes. Direct measurement from dissected eye parts provided refractive index, transmission and reflection. Optical parameters such as absorption and scattering coefficients were calculated with two methods reported in the literature: Kubelka-Munk's (KM) model[28] and Inverse Adding-Doubling (IAD).[29] As another objective, a simulated model – made on COMSOL Multiphysics® – of the eyes is presented based on optical parameters obtained from the measurements.

Different groups have developed simulated models of the human eye. The first model was proposed by Gullstrand.[4] They modelled the cornea, aqueous humour, vitreous humour, retina and lens. The refractive index of the lens is based on a shell model. Subsequently, many models for the same parts have been developed that considerer the accommodation effect[4,5,30] or age dependence.[30,31] The different models deal with the refractive index distribution of the lens according to whether it is a homogeneous distribution,[5,31] a shell model[4] or based on a gradient index.[6] The model developed in this work takes into account the same elements of the eye but also introduce new parts like the iris, the sclera and the ciliary body. The addition of these parts in the simulation model is fundamental in order to develop and improve diffuse optical tomography or transscleral cyclophotocoagulation operations to treat glaucoma.

2. Materials and Experimental Methods

Figure 1 presents the workflow of the experiments described in this paper. Dissection of the eye permitted optical characterization of the various sub-parts of the eye. Extracted refractive index, absorption and scattering coefficients for each part of the eye were then implemented in the simulation model.

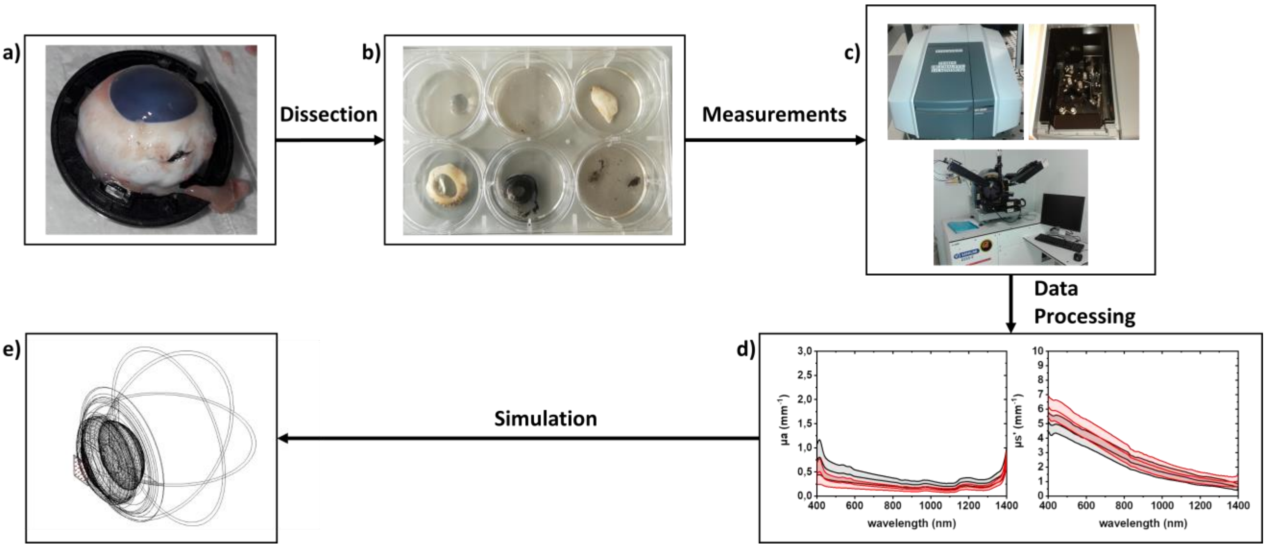


Figure 1: Experimental steps. a) Picture of the eye; b) The different sample of the porcine eye are separated by dissection; c) The reflection and transmission spectrum as well as the refractive index have been measured; d) The absorption and scattering coefficients extracted from the experimental data; e) These coefficients are implemented in COMSOL Multiphysics software in order to create a simulation model of a complete eye

2.1. Theoretical background

Refractive index is a dimensionless quantity that characterizes the light propagation in a medium as well as the change of beam propagation at an interface between two different materials. Absorption coefficient μa (in mm^{-1}) defines the decrease in light intensity (absorption) during propagation of a beam inside a material, while the scattering coefficient μs (in mm^{-1}) defines the change in light direction inside the material [32]. Moreover, the anisotropy coefficient (g) characterizes the direction of the diffused beam in the medium. It is defined by the mean of the cosine of the scattering angle. If $g = 0$ the scattering is isotropic whereas $g<0$ defines a backward scattering and $g>0$ shows a forward scattering[33]. For biological tissues, g ranges from 0.6 to 0.99[3]. Consequently, the scattering is forward. To calculate these three parameters, three measurements are required: the total transmission, the total reflection and the collimated transmission. However, the collimated transmission is not directly measured. We introduce the reduced scattering coefficient (µs') defined by $\mu s' = \mu s \times (1 - g)$. Therefore, we fix g at 0.9 because it is the typical value used for soft tissues [32,34]. The light propagation inside biological tissue can be obtained by solving the radiative transport theory [35]. In order to solve this equation, µa, µs' and g have been calculated using numerical methods.

Kubelka-Munk method

The KM model is a mathematical method for calculating these parameters [28]. It assumes that light propagates through isotropic scattering tissue with a finite thickness and an infinite surface. This implies a large sample in comparison to the size of the light beam in order to neglect the boundary effect. The KM scattering coefficient (S) and KM absorption coefficient (K) have been calculated directly from R, T and the thickness of the sample (d) according to equation 1 and 2.

$$S = \frac{1}{bd} \ln\left(\frac{1 - R(a - b)}{T}\right) \tag{1}$$

$$K = (a - 1)S \tag{2}$$

Where

$$a = \frac{1 + R^2 - T^2}{2R}; b = \sqrt{a^2 - 1} \tag{3}$$

The relationship between S, K and µa, µs' are given by the equation 4 and 5 [36]. The main advantage compared to others methods [37–39] is that these expressions are verified when the incident light is collimated.

$$S = \frac{12}{(4.8446 + 0.472\, g - 0.114\, g^2)^2} \mu s' \tag{4}$$

$$\frac{K}{S} = \frac{\mu a}{\mu s' f(g)} \tag{5}$$

Where

$$f(g) = \frac{6}{(4.8446 + 0.472\, g - 0.114\, g^2)^2} \tag{6}$$

Inverse adding doubling method

IAD [29] is an iterative method. The light propagation in a turbid medium is characterised by three dimensionless parameters: the albedo (α), the optical depth (τ) and g. α and τ are defined as a function of μa and μs in equations 7 and 8.

$$\alpha = \frac{\mu s}{\mu s + \mu a} \tag{7}$$

$$\tau = d(\mu s + \mu a) \tag{8}$$

Moreover, the following assumptions must be verified to apply the IAD method. The measurement is independent of time and is made with unpolarized light. The optical properties of the sample are homogeneous and it has a uniform refractive index. Finally, it is assumed that the element has a finite thickness and an infinite size. We approximate this condition by the fact that the size of the incident beam is fifteen times smaller than the sample.

The IAD method determines the optical parameters for each value of wavelength. The algorithm follows these different steps:

- Starting with a guess of optical parameters
- Calculating the reflection (R_{calc}) and the transmission (T_{calc}) for a thin slab of the sample, then duplicating the results until the sample's thickness is reached.

- Comparing the calculated and experimental data values
- Repeating until calculated and the experimental values match according to equation 9

$$M(\alpha, \tau, g) = \frac{|R_{calc} - R_{meas}|}{R_{meas} + 10^{-6}} + \frac{|T_{calc} - T_{meas}|}{T_{meas} + 10^{-6}} \quad (9)$$

A flowchart explaining the methods is available in the supplementary information (SI Figure 1).

2.2. Eye dissection

A total of 6 porcine eyes were used in these experiments. They were obtained by porcine eye enucleation at the "Centre Européen de Recherche en Imagerie Médicale (CERIMED), Faculté de Médecine" laboratory. Eyes were enucleated directly after the animal death. The porcine age at the time of euthanization was between 5 and 6 months. Pigs from two different breeds were used: Pietrains or Landraces, without any noticeable difference in term of optical parameters.

Immediately after death, the eyes were taken and placed on ice. Transport to the laboratory was completed within two hours after euthanasia. On arrival at the laboratory, the eyes were dissected to extract cornea, lens, vitreous humour, iris, sclera, eyelid, and choroids. Images of the eye are available in the supplementary information (SI figure 2). The samples were placed in a saline solution for a maximum of 24 hours for all the measurements and stored at 4-8 °C in a refrigerator to preserve the tissue.

2.3. Optical measurement on eye parts

The thickness of the samples was measured using the eclipse L200 microscope (Nikon, Japon) connected to the positioner system D35579 WETZRAL (Märzhäuser Wetzlar GmbH & Co, Germany) allowing the measurement of displacement in x, y and z-axis. The sample was deposited on a glass slide. A first focus was made on the glass. Following this, the x, y and z parameters were reset to zero. A second measurement was then taken on the top of the sample. The z-value gave us the thickness of the sample. Thicknesses were measured one day after dissection. The thickness of the vitreous humour was fixed at 1 mm because it was placed between two planar surfaces separated by this distance, due to the soft structure of the vitreous humour.

We measured the reflection and transmission spectrum of each sample using the UV 2600 spectrophotometer (Shimadzu, Japon) equipped with the ISR-2600Plus integrating sphere (Shimadzu, Japon). The data were recorded from 1400 nm to 400 nm with a step of 0.5 nm. The sphere is equipped with two photodetectors, a photomultiplier and an InGaAs detector, so there is a change of detector at 820 nm. The size of the beam was 3×3 mm. These measurements were performed on the same day as the dissection.

To measure the refractive index, we used the GES5E spectroscopic ellipsometry (Semilab, Hungary). The measurements were performed the day after the dissection. The sample was placed on a glass slide and a scan was realized to find the maximum of the reflection at the sample's boundary. Our samples are much thicker than the usual acceptable limit for the ellipsometer. Consequently, we performed the measurement assuming that our sample was infinitely thick and acted as a substrate for the ellipsometer. The measurements were recorded for a range from 400 to 950 nm. We analyse the measurements with the "Spectroscopic Ellipsometry Analyzer" software using a fit based on the Cauchy's law to provide the refractive index.

2.4. Simulation

The simulation model has been realized with the ray optics module of the COMSOL Multiphysics® software. We chose 810 nm for the working wavelength, but it could adjust to any desired wavelength. The reflections were set to the interface with the "specular reflection" wall condition. Concerning the scattering model, we used the diffuse scattering wall condition and inverted the sign of the variable called "reflected ray direction vector, z component". For the reflection and transmission plane, a freeze condition was implemented with an accumulator variable set at each interface to obtain the power received. Finally, for the last simulation with the complete eye, the refractive index and the absorption are set for all the different parts. Furthermore, scattering was added for ciliary body, iris and sclera. Concerning the sclera, the reflection was also taken into account. The freeze condition was put in the retina surface to stop the light propagation.

3. Experiments and Results

Figure 2 presents the refractive index results obtained for wavelengths ranging from 400 to 950 nm. As expected, and following the Cauchy experimental expression, the refractive index decreases when the wavelength increases. The results were obtained from an average of at least three different samples for each element. The R^2 of each fit was greater than 0.92. Table 1 presents our results at 650 nm compared to the results found in the literature. Given the dispersion of reported refractive indices, our measurements were consistent with those reported in the literature for the lens, the vitreous humour and the sclera. The refractive index of the lens measured in our case corresponded to an equivalent mean value. However, a difference is noticeable between the refractive index of the lens for humans and pigs. Concerning the iris and the choroids, we did not find any information in the literature; barring any comparison. We did not obtain conclusive values for the cornea due to its rounded structure, nor for the eyelid due to its rough surface, which seriously altered the quality of the measurements. In fact, the planarity of the sample is an essential feature to measure the refractive index with the ellipsometry technique. The reported refractive indices were equal to 1.376*[12]* for the cornea and 1.395*[40]* for the eyelid, which corresponds to the refractive index of the porcine epidermis.

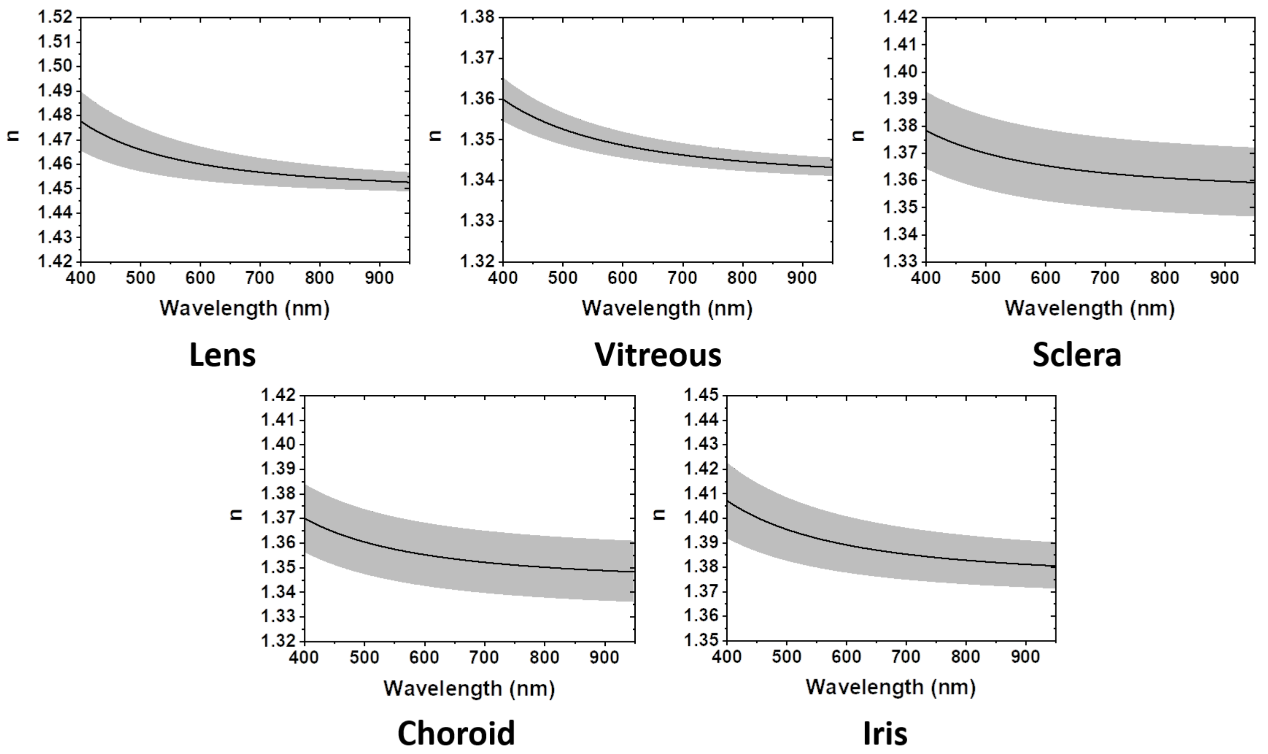


Figure 1: Solid curve shows the average refractive index for porcine eyes versus the wavelength ranging from 400 to 950 nm measured with the GES5E spectroscopic ellipsometry (Semilab, Hungary) from at least 3 samples. The fit is based on the Cauchy's law and the R^2 for the data are at least equal to 0.92

Table 1: Average and standard deviation of the refractive index obtained for the different samples of the porcine eyes at 650 nm measured with the GES5E spectroscopic ellipsometry (Semilab, Hungary) from at least 3 samples. The values found in the literature for the same sample are also presented.

Sample	**Refractive index @650 nm**	**Literature**	
		Refractive index	Comments
Cornea		1.376[12]	mean value of human refractive index
Lens	1.458 ± 0.006	1.4955[7]	@633 nm, porcine
		1.4686[8]	@633 nm, porcine
		1.449[9]	@633 nm, porcine
Iris	1.387 ± 0.011	no data available	

Sclera	1.364 ± 0.013	1.42[13]	calculated from an average between that of collagen (n=1.47) and of the fibrils (n=1.36)
		1.385 ± 0.005[14,41]	@589 nm, human
Vitreous humour	1.341 ± 0.009	1.3322[10]	@633 nm, porcine
		1.351[11]	@654 nm, bovine
		1.336[12]	@589 nm, human
Choroids	1.354 ± 0.013	no data available	
Eyelids		1.395[40]	@650 nm, porcine epidermis

From the transmission and reflection measurements of each part of the eye, we could calculate μa and μs' (in mm^{-1}). The refractive indices presented above are used in the IAD method. Figure 2 and Figure 3 present the absorption and reduce scattering coefficients versus the wavelength, respectively, calculated by the KM and IAD methods from the experimental data. The transmission and reflection spectrum (raw data) as a function of the wavelength are presented in the supplementary information (SI figure 3). The sample thickness used for both methods are summarized in table 2. We can notice some differences with the thickness measurements of samples. For example for the cornea, previous studies have shown lower thicknesses [27,42,43]. This can be explained by the age and breed of the pig but also by the swelling effect due to storage in saline solution. We observed three different trends. First, the cornea, lens and vitreous humour have low absorption coefficients. They allow light to pass through ocular media without attenuation. This is the normal light path for vision. The second group of elements comprise the eyelid and the sclera. They have a higher absorption coefficient. They protect and give structure to the eye. The last group is composed of choroids and iris. They absorb the light. The iris controls the amount of light that enters into the eye whereas choroids absorb light at the back of the eye. The last two groups absorb more in the visible region than in the near-infrared. Peaks at 980, 1200 and 1400 nm are due to water absorption [44,45].

Table 2: Thickness of the samples measured with the eclipse L200 microscope (Nikon, Japon) connected to the positioner system D35579 WETZRAL (Märzhäuser Wetzlar GmbH & Co, Germany). The data are the average and the standard deviation of at least 4 samples

Sample	Thickness (mm)
Cornea	1.73 ± 0.47
Lens	5.28 ± 0.90
Iris	1.13 ± 0.29
Sclera	1.80 ± 0.42
Vitreous humour	1.00
Choroids	0.66 ± 0.26
Eyelids	3.49 ± 0.79

From Figure 3, we notice that for the cornea, the lens, the vitreous humour, the sclera and the eyelid, scattering predominates over absorption. The scattering coefficient is low for the cornea, lens and vitreous humour. As for the absorption, these results are coherent in order to allow the convergence of the rays on the retina. For choroids and iris, the absorption is more substantial than the scattering effect but remains an important phenomenon for light propagation in those samples.

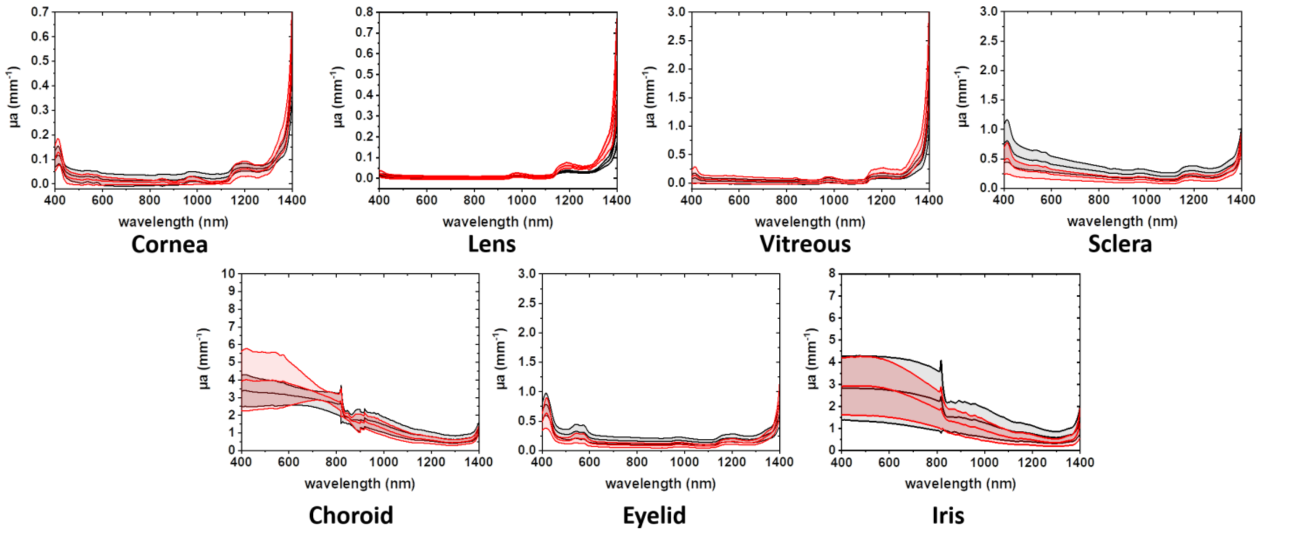


Figure 2: Average absorption coefficient for porcine eyes versus the wavelength ranging from 400 to 1400 calculated from the experimental data by the KM (black) and IAD (red) methods. The data are extracted from at least 4 samples.

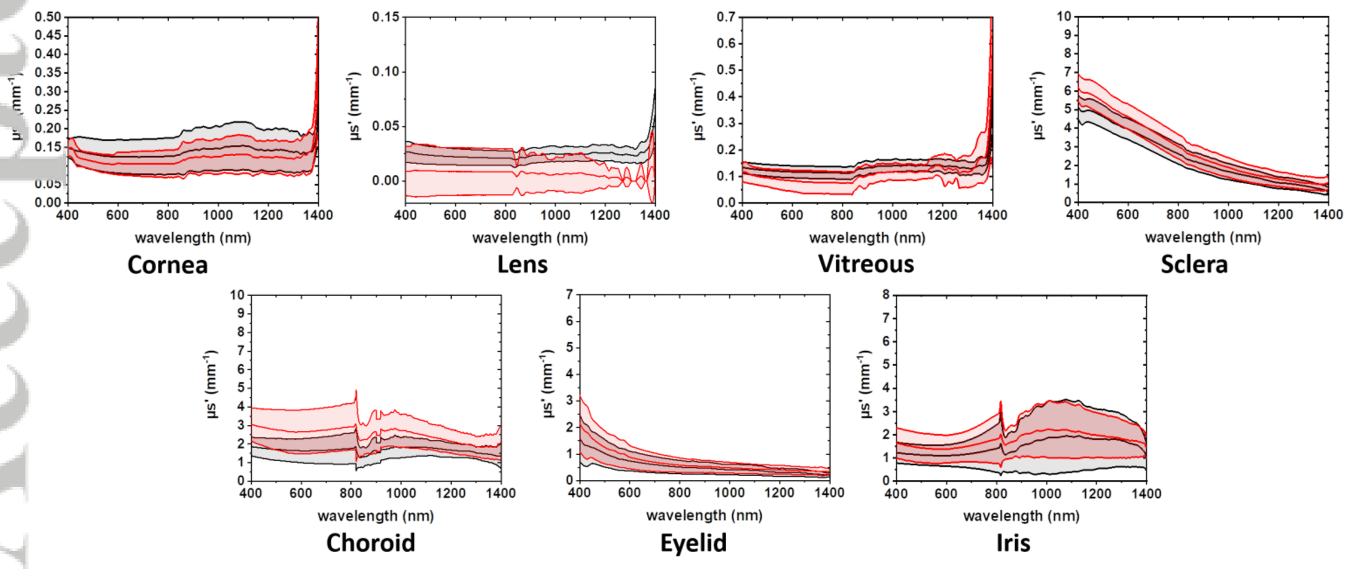


Figure 3: Average reduce scattering coefficient for porcine eyes versus the wavelength calculated from the experimental data by the KM (black) and IAD (red) methods. The data are extracted from at least 4 samples.

During transscleral cyclophotocoagulation surgery, laser diodes of this wavelength are used because they are small, portable and ciliary body absorbs more at 810 nm and therefore causes more focused destruction and decrease the inflammation [46,47]. Table 3 compares our result with the absorption and scattering coefficients found in the literature at 810 nm. Overall obtained results show a good consistency with those of the literature. Notably, the optical coefficients parameters remain in close range. However, since experiments were on different animal species with different experimental conditions it is delicate to make a direct comparison. Extracted from our experiment, scattering coefficients are higher than absorption for the cornea, lens and vitreous humour. However, the absorption and scattering coefficients for these parts are at least an order of magnitude lower than those of which were obtained from the sclera, the iris, the choroids and the eyelid. As expected, the iris has a very high absorption coefficient because it controls the amount of light entering the eye and therefore the depth of field. The μa and μs' values for the cornea, lens, vitreous humour and sclera are of the same order of magnitude as the literature and follow the same trend. Nevertheless, scattering coefficients for cornea and vitreous humour are an order of magnitude larger than those found in other works. Literature results are not obtained on porcine but on bovine or human eye; which could explain the disparity. Even though, the optical parameters of the iris and choroids are poorly detailed in the literature or with different measurement protocols [23], the curves of choroids follow the same trend. However, for the iris due to the disease and the study of the stroma alone, the published results on the absorption and reduced scattering coefficient are lower than ours.[26] For choroids, as the comments in the table show, the measurement protocol is different, which explains the differences observed for μa and μs'. Concerning the eyelid, μa and μs' decrease towards the infrared. This is in agreement with the literature where studies demonstrated the infrared filtering properties of eyelids.[17,48] We implement all the parameters presented here in a simulation model in order to understand the propagation of light in the eye.

Table 3: Comparison between absorption and scattering coefficient determined with KM and IAD methods obtained from the processing of porcine eye raw data. A comparison with the values found in the literature is performed. The comments column presents the animal species of the study as well as the wavelength of the measurement. The data are the average and the standard deviation of at least 4 samples.

Sample	**KM**		**IAD**		**Literature**		
	μa (mm^{-1})	μs' (mm^{-1})	μa (mm^{-1})	μs' (mm^{-1})	μa (mm^{-1})	μs' (mm^{-1})	comments
Cornea	0.014 ± 0.02	0.126 ± 0.049	0.006 ± 0.009	0.106 ± 0.037	0.0022	[0.0016 , 0.0234] for g = [0.85 , 0.99]	@800 nm, bovine, 2 eyes [21]
Lens	0.003 ± 0.006	0.021 ± 0.006	0.003 ± 0.007	0.009 ± 0.021	0.0005	[0.0023 , 0.0338] for g = [0.85 , 0.99]	@800 nm, bovine, 2 eyes [21]
Iris	2.234 ± 1.341	1.438 ± 1.071	1.842 ± 0.817	1.843 ± 1.088	0.290	4.003	@750 nm, vitiligo rabbit, 2 eyes [26] Study of stroma (majority component of the iris)
Sclera	0.284 ± 0.085	2.558 ± 0.539	0.164 ± 0.052	2.936 ± 0.654	0.16 ± 0.05 0.0608 0.1425	2.983 ± 0.75 3.604 3.49	@850 nm, rabbit, eye of four rabbit [13] @810 nm, human [14] @810 nm, porcine, 4 eyes[24]
Vitreous humour	0.014 ± 0.038	0.121 ± 0.015	0.033 ± 0.048	0.057 ± 0.035	0.105 0.048	0.0004 0.0003	@980 nm, bovine, 2 eyes [11] @980 nm, human, 2 eyes[11]
Choroids	2.620 ± 0.618	1.771 ± 0.839	2.759 ± 0.429	2.940 ± 1.250	8.456	8.000	@810, human, 2 or 3 eyes[23] Protocol: •Dissection •Freezing the samples with liquid propane •Pulverizing them to obtain a homogeneous analysis surface.
Eyelids	0.161 ± 0.055	0.503 ± 0.249	0.088 ± 0.040	0.574 ± 0.268	No data found		Only transmission and reflection spectrum

4. Simulation

All the simulated models from the literature presented in the introduction only considered the cornea, the lens, the pupil, the aqueous / vitreous humour and the retina. However, some laser surgeries for glaucoma treating require light to pass through the sclera.[46] We propose a model to simulate at the same time the different elements involved in the visual path, but also including the iris, the sclera and the ciliary body. Our model is defined for wavelengths from 400 to 1400 nm. The wavelength arbitrarily chosen for the simulation was 810 nm for the same reason presented above.

We used the Ray Optics module available in COMSOL Multiphysics® to perform these simulations. It was carried out in 2 steps. First, we validated the model by performing simulations of each sample separately in order to compare the transmission and reflection values obtained in the simulation with the measured one. After validations of each part, we implemented these values in a 3D version of the eye.

Figure 5.a shows the simulation of individual eye parts in comparison with transmission obtained by spectrophotometric experiments. One light source (LED) and two photodetectors – for reflection (for the top plane) and transmission (for the bottom plane) – complete the simulation. To validate our model, Table 1 in supporting information presents the different parameters used; they are extracted from the previous experiments. Moreover, we set the value of the reflection at the top of the sample boundary according to the measured values. The aqueous parameters have been taken as equal to the vitreous parameters. The simulated reflection and transmission at the detectors were calculated according to equations 10 and 11, respectively.

$$R_{sim} = \frac{Power\ received\ at\ the\ reflection\ photodetector}{Initial\ power} \times 100 \tag{10}$$

$$T_{sim} = \frac{Power\ received\ at\ the\ transmission\ photodetector}{Initial\ power} \times 100 \tag{11}$$

The COMSOL Multiphyics® software allows the setting of specular, diffuse reflection and transmission according to the beer lambert's law at the interface of a material. It also takes into account the absorption of the material but does not propose a model for scattering transmission. We have developed our own scattering transmission model. For this, multiple planes emulate the scattering transmission in the dedicated media. The number and the space of them are a function of two parameters: the thickness of the material noted d and the reduced mean free path (ls'). The term ls' corresponds to the theoretical distance that a photon travels between two scatters. It is directly related to µs' by equation 12. We defined the number of planes (np), corresponding to the number of theoretical scattering events inside the sample, as defined by equation 13. Here, np is rounded up to the nearest whole number. The percentage of scattering (%s) at each interface was calculated by the ratio between ${}^{d}/_{ls'}$ and np as we see in the equation 14. This parameter is set at the plane's boundary conditions allowing control of the percentage of rays that must be scatted at each plane. The distance between each plane is equal to ls'.

$$ls' = \frac{1}{\mu s'} \tag{12}$$

$$np = quotient\left(\frac{d}{ls'}\right) + 1 \tag{13}$$

$$\%s = \frac{{}^{d}/_{ls'}}{np} \tag{14}$$

Figure 5.b presents a 2D simulation of this scattering phenomenon. With the same probability of diffusion at each interface fixed at 10%, the direction of two rays changes at each interface. The scattering transmission model can be adjusted to all situations due to np, %s variable. It makes it possible to model both low and high scattering materials. We present two particular cases, the lens, which is characterized by a weak scattering coefficient, and the sclera, which is a diffusing part of the eye, as we have seen in the experimental study, in order to test the accuracy of the scattering model. Two different simulations have been performed for these two elements, one taking into account only the absorption and a second also taking into account the scattering (Figure 5.c). Simulation plots for both cases are presented in the supplementary information (SI, figure 4)

For the lens, neglecting diffusion does not affect the simulation value. It is slightly lower because the software reduces the light power when the rays pass through an optical index change interface even if the absorption coefficient is set to zero and the rays arrive perpendicular to the interface. Moreover, we notice that the simulated value with the diffusion is lower than the one obtained without. This has the consequence of increasing the gap with the experimental result. For sclera, we see that the results are closer to the experimental value when diffusion is taken into account. This result confirms that sclera is more diffusive than absorbent. Although the results obtained for the sclera show similar results for both analysis methods, we note that the gap between the measured value and simulated result obtained from KM coefficient (3.52 %) was smaller than that using IAD (11.45 %). We observe the same tendency for the other parts of the eye, so for the next simulations, we consider only the result with KM.

Figure 5.d shows the results for the lens and sclera depending on whether reflection at the material

interface was considered or not. The scattering phenomenon is considered only for the sclera in view of the previous simulation. In both cases, neglecting the reflection at the interface increases the transmission. This increase is attributed to the absence of reflected rays. If the reflection is considered, some of the incident rays are reflected. As a result, the amount of light entering inside the eye is reduced. By contrast, when the reflection is neglected, all the rays enter inside the eye. For the lens, this phenomenon is not detrimental for our work because the result is closer to the experimental value than the simulation results with the reflection. On the other hand, for the sclera, this phenomenon causes large discrepancy because it leads to a sizeable error in simulation. For this reason, the reflection was ignored in the simulated model for the lens but not for the sclera. The simulation performed on all the parts is summarized in the supplementary information (SI figure 5).

Correspondingly, we generalize the previous simplifications. The scattering effect was considered only for sclera, choroids, iris and eyelid, which have a high scattering coefficient. Similarly, the reflection was only applied at the interface of the sclera and eyelid because for all other samples the value is weak which induces a negligible error (Figure 5.e). The largest difference between the experimental and simulated value was obtained for choroids with a gap of 5.29%. We also neglected the absorption for the vitreous and aqueous humour because after simulations, we concluded that we needed to ignore this phenomenon to match the experimental data. We observed that the results obtained in the simulation were consistent with those measured.

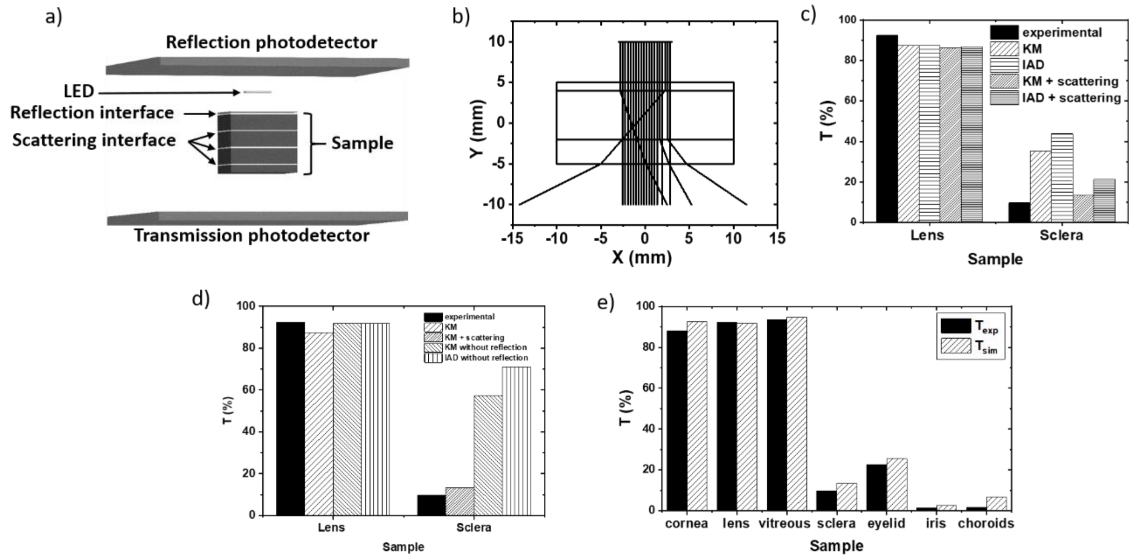


Figure 4: Simulation on Comsol a) Model used to calculate the transmission and the reflection of the eye sample; b) 2D simulation to explain the scattering principle. The media absorption is set to zero. Diffusion at each interface is fixed at 10% c) Influence of the scattering effect for the lens and the sclera. Comparison between the KM and IAD methods used for the analyses; d) Influence of the reflection at the interface for the lens and the sclera; e) Comparison between results obtained on simulation and the experimental data for the different parts of the porcine eye

Figure 6 shows different simulations of the complete eye. Figures 6.a and 6.b represent 2D and 3D simulations, respectively, of light propagation from cornea to retina. We thus simulated the amount of light that reached the retina when the eye is open. The light passes through the different medium with a small absorption. The percentage of light received by the retina is 95% in both the simulations which is in agreement with the literature [18]. A video illustrating this phenomenon is available in the supplementary information (SI figure 6). Similarly, figures 6.c and 6.d show the case of non-standard propagation of the sclera towards the retina. In this case, the scattering phenomenon is important and the ciliary bodies absorb the majority of the light. Our simulation confirmed that the ciliary bodies absorb 82% and 90.5% of the IR light for the 2D and 3D model, respectively, which by heating allows their destruction, as already used in the treatment of glaucoma. These results are in agreement with the experimental data of choroids absorption. Moreover, the ciliary body model is based on choroids measurement because the absorption and scattering coefficients for ciliary body and choroids found in the literature present similar values.[12] During the transscleral cyclophotocoagulation, surgeons destroy a part of the ciliary bodies and thus release the pressure inside the aqueous chamber by the IR illumination. A video presenting the propagation of light from sclera to retina is presented in the supplementary information (SI figure 7).

This type of simulation is useful to improve the efficiency of surgery. In this case, the retina receives only 6% and 3% of the light for the 2D and 3D simulations. In the 2D simulation, the propagation is done according to x and y axes, while for the 3D simulation, it is done according to x, y and z axes. As a result, the distance travelled in absorbent media is longer for 3D simulations. This explains the differences in the absorption values observed between the simulations.

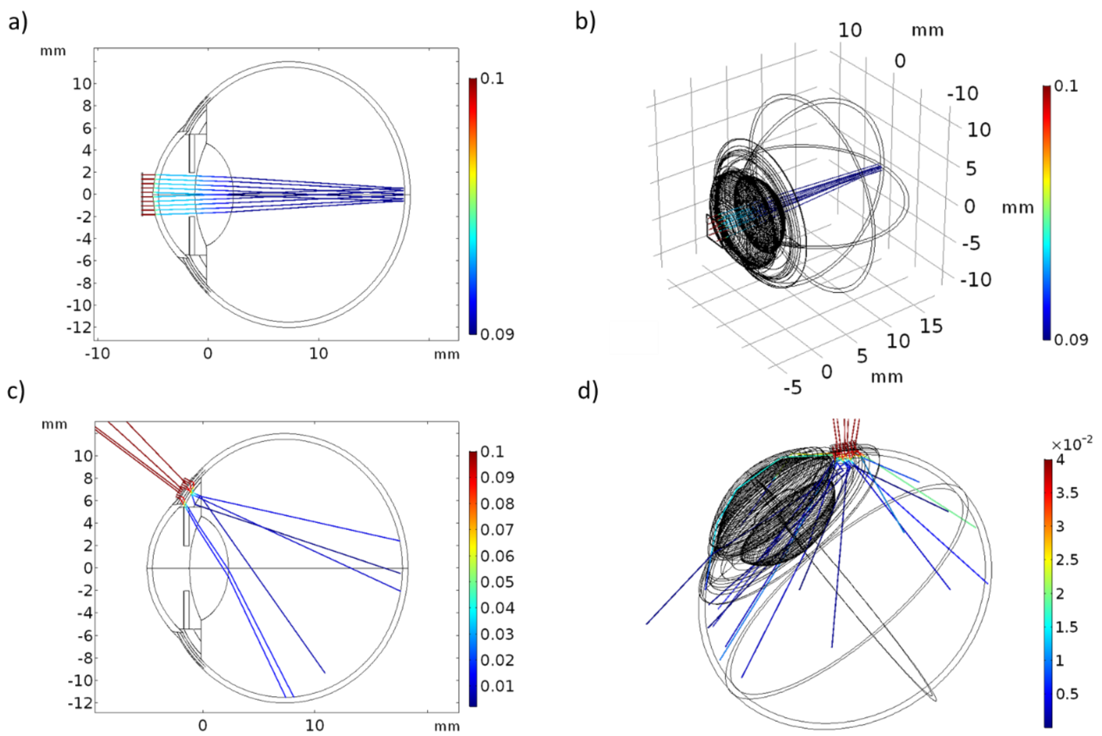


Figure 5: Simulation of the complete eye on Comsol a) 2D simulation for the convergence of the light from the cornea to the retina; b) 3D simulation for the convergence of the light from the cornea to the retina; c) 2D simulation for non-standard light propagation inside the eye; d) 3D simulation for non-standard light propagation inside the eye. For all graphs, the colour gradient represents the ray power in watts with blue corresponding to low power and red to high power.

5. Conclusion

This work gathers the optical parameters of the different parts of the porcine eye for wavelengths between 400 nm to 1400 nm. The study was conducted on the cornea, lens, vitreous humour, sclera, iris, choroids and eyelid. We measured the refractive index as well as the transmission and reflection for each sample. Moreover, the refractive index is measured of wavelengths from 400 to 950 nm. These results complement those found in the literature and provide values for iris and choroids. The absorption and scattering coefficients were calculated from the KM and IAD methods. The results were obtained under the same test conditions and are coherent with the available data from the literature. To the best of our knowledge, this is the first characterization study in the visible and infrared region using these seven elements of the eye in the same experimental conditions. In addition of measuring these parameters, a simulation was implemented in order to develop a complete eye model as realistic as possible. A scattering transmission model has been developed to improve the accuracy of the simulation. Each sample has been tested separately with the aim of validating the simulated results. The developed model showed consistent data because the biggest difference between simulated and experimental values was as low as 5.29%. The global simulation provides results in agreement with the literature data. This comprehensive model advances our understanding of the interaction of light within the eye and should thus facilitate the development of new surgical procedures using lasers. This model gives the possibility to test different procedures associated to light surgeries. The model allows quantifying the percentage of light absorbed for each part of the eye and seeing on which part it is likely to have heat elevations. COMSOL is a multiphysics software where temperature sensors or where mechanical characterization could be easily implemented. Thanks to the simulation, ophthalmologists will be able to test different solutions such as wavelength, light power and quantify undesirable effects. They will also be able to choose the best light sources placement and the intensity required before conducting clinical tests. Indeed, systematic and complete optical characterization and simulation of light propagation into eye conducted in this work represent a valuable piece of information for ophthalmologic clinical use.

References

[1] Ocular Surgery News U.S. Edition, April 10, 2019, " Laser refractive surgery gradually regains ground since peaking in early 2000s"

[2] Remington, L. A., (Elsevier/Butterworth Heinemann, St. Louis, Mo 2012).

[3] Jacques, S. L., Phys. Med. Biol.,**58**, R37, (2013).

[4] Gullstrand, Helmholtz’s Handbuch Der Physiologischen Optik,. 1909, p. Appendices II and IV.

[5] Navarro, R., J. Santamaría, and J. Bescós, J. Opt. Soc. Am. A, JOSAA,**2**, 1273, (1985).

[6] Liou, H.-L., and N. A. Brennan, J. Opt. Soc. Am. A, JOSAA,**14**, 1684, (1997).

[7] Vilupuru, A. S., and A. Glasser, Ophthalmic and Physiological Optics,**21**, 296, (2001).

[8] Wong, K.-H., S. A. Koopmans, T. Terwee, and A. C. Kooijman, Invest. Ophthalmol. Vis. Sci.,**48**, 1261, (2007).

[9] Birkenfeld, J., A. de Castro, S. Ortiz, D. Pascual, and S. Marcos, Vision Research,**86**, 27, (2013).

[10] Sivak, J. G., and T. Mandelman, Vision Research,**22**, 997, (1982).

[11] Sardar, D. K., G.-Y. Swanland, R. M. Yow, R. J. Thomas, and A. T. C. Tsin, Lasers in Medical Science,**22**, 46, (2007).

[12] Atchison, D. A., and G. Smith, (Butterworth-Heinemann, Oxford 2002).

[13] Nemati, B., H. G. Rylander, and A. J. Welch, Appl. Opt., AO,**35**, 3321, (1996).

[14] Bashkatov, A. N., E. A. Genina, V. I. Kochubey, and V. V. Tuchin, Optics and Spectroscopy,**109**, 197, (2010).

[15] Boettner, E. A., and J. R. Wolter, Investigative ophthalmology & visual science,**1**, 776, (1962).

[16] Geeraets, W. J., R. C. Williams, G. Chan, W. T. Ham, D. Guerry, and F. H. Schmidt, Arch Ophthalmol,**64**, 606, (1960).

[17] Robinson, J., S. C. Bayliss, and A. R. Fielder, Vision Res.,**31**, 1837, (1991).

[18] Geeraets, W. J., and E. R. Berry, American journal of ophthalmology,**66**, 15, (1968).

[19] Watts, G. K., The British journal of ophthalmology,**55**, 60, (1971).

[20] Van Norren, D., and L. F. Tiemeijer, Vision Res.,**26**, 313, (1986).

[21] Yust, B. G., L. C. Mimun, and D. K. Sardar, Lasers in Medical Science,**27**, 413, (2012).

[22] Sardar, D. K., B. G. Yust, F. J. Barrera, L. C. Mimun, and A. T. C. Tsin, Lasers in Medical Science,**24**, 839, (2009).

[23] Hammer, M., A. Roggan, D. Schweitzer, and G. Muller, Physics in medicine and biology,**40**, 963, (1995).

[24] Chan, E. K., B. Sorg, D. Protsenko, M. O’Neil, M. Motamedi, and A. J. Welch, IEEE Journal of selected topics in quantum electronics,**2**, 943, (1996).

[25] Vogel, A., C. Dlugos, R. Nuffer, and R. Birngruber, Lasers in surgery and medicine,**11**, 331, (1991).

[26] Koblova, E. V., A. N. Bashkatov, L. E. Dolotov, Y. P. Sinichkin, T. G. Kamenskikh, E. A. Genina, and V. V. Tuchin, in: Tuchin, V. V. (Ed.). 2007, pp. 653521-653521–10.

[27] Sanchez, I., R. Martin, F. Ussa, and I. Fernandez-Bueno, Graefe’s Archive for Clinical and Experimental Ophthalmology,**249**, 475, (2011).
[28] Kubelka, P., J. Opt. Soc. Am., JOSA,**38**, 448, (1948).
[29] Prahl, S. A., M. J. C. van Gemert, and A. J. Welch, Appl. Opt., AO,**32**, 559, (1993).
[30] Coelho, J. M. P., A. Baião, and P. Vieira, in: Martins Costa, M. F. P. C. (Ed.). 2013, p. 8785CS.
[31] Norrby, S., Ophthalmic Physiol Opt,**25**, 153, (2005).
[32] Welch, A. J., and M. J. C. Gemert, (Springer US : Imprint : Springer, Boston, MA 1995).
[33] Cheong, W.-F., S. A. Prahl, and A. J. Welch, IEEE journal of quantum electronics,**26**, 2166, (1990).
[34] Tuchin, V. V., (SPIE/International Society for Optical Engineering, Bellingham, Wash 2007).
[35] Chandrasekhar, S., (Dover Publications, New York, NY 1960).
[36] Thennadil, S. N., JOSA A,**25**, 1480, (2008).
[37] Brinkworth, B. J., Appl Opt,**11**, 1434, (1972).
[38] Gate, L. F., Appl Opt,**13**, 236, (1974).
[39] Star, W. M., J. P. A. Marijnissen, and M. J. C. van Gemert, Physics in Medicine and Biology,**33**, 437, (1988).
[40] Ding, H., J. Q. Lu, K. M. Jacobs, and X.-H. Hu, J. Opt. Soc. Am. A, JOSAA,**22**, 1151, (2005).
[41] Tuchin, V. V., I. L. Maksimova, D. A. Zimnyakov, I. L. Kon, A. H. Mavlyutov, and A. A. Mishin, JBO, JBOPFO,**2**, 401, (1997).
[42] Jay, L., A. Brocas, K. Singh, J.-C. Kieffer, I. Brunette, and T. Ozaki, Opt. Express, OE,**16**, 16284, (2008).
[43] Faber, C., E. Scherfig, J. U. Prause, and K. E. Sørensen, Scand. J. Lab. Anim. Sci.,**35**, 6, (2008).
[44] Hale, G. M., and M. R. Querry, Appl. Opt., AO,**12**, 555, (1973).
[45] Kou, L., D. Labrie, and P. Chylek, APPLIED OPTICS,**32**, 3531, (1993).
[46] Gupta, N., and R. N. Weinreb, J. Glaucoma,**6**, 426, (1997).
[47] Pastor, S. A., K. Singh, D. A. Lee, M. S. Juzych, S. C. Lin, P. A. Netland, and N. T. A. Nguyen, Ophthalmology,**108**, 2130, (2001).
[48] Bierman, A., M. G. Figueiro, and M. S. Rea, Journal of biomedical optics,**16**, 067011, (2011).

Graphical Abstract for Table of Contents

Eye simulation models allow the development of new laser treatments in a simple and safe way for patients. To this purpose, the creation of an eye simulated model based on optical parameters obtained from experimental data is presented in this study. This model will facilitate the understanding of the light propagation inside the porcine eye